
ISyE 4600 - Fall 2025
Project Proposal

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Project Title: Hybrid-ML Digital Twin for Real-Time Yield and in
Biopharmaceuticals

Abstract

This project proposes a Hybrid-ML Digital Twin framework designed to bridge the gap between the complex biological system that the pure mechanistic modeling and pure machine learning modeling struggles to interpret and predicts the critical quality attributes (e.g., antibody yield) from bioreactors. [1][2][3] The Directed acyclic graph successfully gives the framework of the feature relationship to downstream prediction [4] of the antibody yield from the hybrid model. The hybrid model consists of the titer prediction constructed from mechanistic equation and residual from XGBoost prediction. [5][6] By integrating mechanistic Monod kinetics with stochastic machine learning forecasting, the system enables proactive and precise prediction of the antibody yield. The framework focuses on minimizing error across predicted yield distributions over days, effectively reduces the bias from conventional biopharmaceutical manufacturing process analysis. [3]

1 Introduction

Challenges in Bioprocess Digitalization

Biopharmaceutical manufacturing is currently navigating a “Data-Rich, Information-Poor” paradox. While bioreactors generate high-dimensional sensor data, the biological stochasticity and complex metabolic interactions make accurate real-time yield prediction exceptionally difficult. Although computational methods have revolutionized protein structure prediction and functional genomics, AI-powered advancements in downstream manufacturing processes and real-time quality control remain nascent [1].

The Mechanistic vs. Data-Driven Gap

Traditional mechanistic models, such as Monod kinetics, provide high interpretability and adhere to fundamental mass balances; however, they often fail to account for the stochastic noise and batch-to-batch variability found in living systems. Conversely, pure Machine Learning (ML) models often perform poorly in bioprocessing because they lack physical grounding, leading to “black-box” predictions that may violate thermodynamic constraints in the biological system [2]. Consequently, a Hybrid-ML approach is optimal.

Current hybrid model black-box

The current hybrid models utilize different combination of the mechanistic/physical model and soft-sensing machine learning model. This approach moved beyond conventional modeling or multivariate correlation analysis [3]. There are two main frameworks that consider kinetic equation into the ML framework: (1) Integrating kinetic constraint into the ML layers [6] and (2) Using calculated result from kinetic model and adding trained ODE residual [5]. However, the feature relationships of the input parameters aren't considered before training, potentially causing the bias in extracting true relationship between causal effect for the critical quality attributes. [4]

Related work

The data of the project refers to the research predicting titer, viable cell density (VCD), and cumulative glucose consumption which utilized different ML to predict specific rate before calculating on mechanistic equation [5]. PINN is one of the Scientific Machine Learning (SciML) [6], which integrated the mechanistic restriction (e.g., Monod equation) to make sure the ML model satisfy the kinetic characteristics of the biological features.

2. Data

The dataset is sourced from Yatipanthalawa et al. [5], comprising bioreactor process data from a mammalian cell culture system, and includes 29 bioreactors and 379 experimental data of 0-12 days for each reactor. The feature is shown in table 1:

Table 1 Feature table for the dataset

Reactor	Working_Day	Volume	Glucose Feed	VCD	Titre	Temperature	Glucose concentration	Glucose consumed	Cumulative Glucose consumed	Lactate concentration	Protein amount
1	0	0.52227	0	0.0159	0.00095	0.975609756	0.927107062	0	0	0.025641026	0.000929914
1	1	0.50797	0	0.05893	0.00897	0.975609756	0.851936219	0.125485194	0.02237317	0.111111111	0.007579379
1	2	0.49366	0	0.16946	0.02315	0.975609756	0.67881549	0.267327814	0.070035931	0.427350427	0.01922668
1	3	0.4794	0	0.36477	0.0678	1	0.412300683	0.399224813	0.141215043	0.743589744	0.055716738
Viability		Specific growth rate	Specific productivity		Specific cumulative gluc rate						
0.90506329		6.353626723	0.91545819		1.081451445						
0.96202532		6.522305938	0.927726665		0.733799277						
0.94303798		5.634815818	0.957389408		0.418686571						
0.94620253		3.216081333	1.016053169		0.261899279						

3. Methodology

This project leverages the integration of biological causal effects, mechanistic modeling, and machine learning. The causal relationships among input features—such as viable cell density (VCD), titer, temperature, and growth rate—are represented via a directed acyclic graph (DAG) and integrated into the mechanistic model. This mechanically predicted titer is further subtracted by the actual yield from dataset, whose residual yield ΔT is as:

$$\Delta T = T_{\text{mech}} - T_{\text{actual}}$$

This residual, along with the DAG structure, serves as the training input for XGBoost. After training, the final prediction of the hybrid model is defined as:

$$\hat{T}_{\text{hybrid}} = T_{\text{mech}} + \hat{\epsilon}_{\text{XGB}}$$

Further evaluation and comparison of pure mechanistic model, hybrid-ML model, and pure ML are demonstrated in 4.1 and 4.2.

3.1 Directed acyclic graph (DAG) layer

DAG for Bioreactor Parameter Analysis in Protein Production

DAG is a method that assigns the cause-and-effect relationship between different nodes and edges, helps identify the confounding variables that require conditioning while building DAG. In this project, DAG is applied in bioreactor parameters to demonstrate the flow of the biological reaction for protein production. The causal edge definitions are the features from the data set; its direction is defined by (1) literature (2) Stoichiometry (3) NOTEARS algorithm [19].

(1) Literature-Required Edges

These edges are corresponded to established mechanistic laws.

Monod Kinetics

- **Glucose_Feed $\rightarrow$ Glucose_conc**
Fed-batch mass balance: $dS/dt = F \cdot S_F/V - \mu X/Y_{X/S}$. Each glucose feed directly raises dissolved concentration.
- **Glucose_conc $\rightarrow$ μ (specific growth rate)**
Monod equation: $\mu = \mu_{\text{max}} \cdot S / (K_s + S)$. The specific growth rate is a saturating function of substrate concentration.[7]
- **Temperature $\rightarrow$ μ (specific growth rate)**
Temperature modulates μ through an Arrhenius-type relationship: $\mu(T) \propto \exp(-E_a / RT)$. Temperature shifts are a standard PAT strategy to switch CHO cultures from growth to production phase.[9][10]

Cell Mass Balance

- **$\mu \rightarrow$ VCD (Viable Cell Density)**

Biomass balance: $dX/dt = \mu X - DX$. Growth rate is the sole positive driver of cell density increase in a fed-batch with no cell bleed.[11]

Growth-Associated Specific Rates (Luedeking-Piret)

- **$\mu \rightarrow q_p$ (production rate/cell)**
Luedeking-Piret model: $q_p = \alpha \cdot \mu + \beta$. The growth-associated term ($\alpha \cdot \mu$) links productivity to growth rate. [8][12]
- **$\mu \rightarrow q_s$ (glucose uptake rates)**
Stoichiometric linkage: $q_s = \mu / Y_{X/S} + m_s$, where m_s is maintenance energy. Higher growth rates require higher glucose uptake rates. [13]

(2) Substrate Consumption Stoichiometry

- **VCD $\rightarrow$ Glucose_consumed**
Absolute consumption rate = $q_s \cdot \text{VCD}$. More viable cells = more glucose consumed per time step.
- **Glucose_consumed $\rightarrow$ Cumul_Glucose**
Cumulative consumption is the running integral of per-step consumption; a purely deterministic relationship.
- **Protein_amount $\rightarrow$ Titre**
Titre is defined as extracellular protein concentration = $\text{Protein_amount} / \text{Volume}$. [18]

Aerobic Glycolysis (Warburg-like Effect in CHO)

- **VCD $\rightarrow$ Lactate_conc**
Each viable cell secretes lactate as a byproduct of aerobic glycolysis (high glycolytic flux even under aerobic conditions, analogous to the Warburg effect). [14][15]
- **Glucose_consumed $\rightarrow$ Lactate_conc**
Stoichiometrically, lactate yield from glucose is approximately 1.3–1.8 mol/mol in CHO, depending on metabolic state.[14]

Cell Viability Inhibition

- **Lactate_conc $\rightarrow$ Viability**
High lactate concentrations directly inhibit cell growth and increase apoptosis. The inhibitory effect follows an uncompetitive or non-competitive inhibition model. [16][17]
Sub-optimal temperatures ($>37^\circ\text{C}$ or temperature shifts that are too abrupt) reduce the live cell fraction by inducing heat-shock responses and apoptosis. [10]

Product Accumulation

- **VCD $\rightarrow$ Protein_amount**
Total secreted protein = $\int q_p \cdot \text{VCD} dt$. Higher cell density directly scales product accumulation. [12]
- **$q_p \rightarrow$ Protein_amount**
Higher per-cell productivity directly increases accumulation rate independent of cell count. [8]

(3) NOTEARS-Learned Edges

These are edges discovered by the NOTEARS algorithm [19] from the observational data that were not explicitly required by the literature skeleton. They are kept if they are not physically forbidden and do not create cycles.

DAG result

Based on the literature reviews, the design of DAG for bioreactor reaction is demonstrated in the following figure 1 and figure 2.

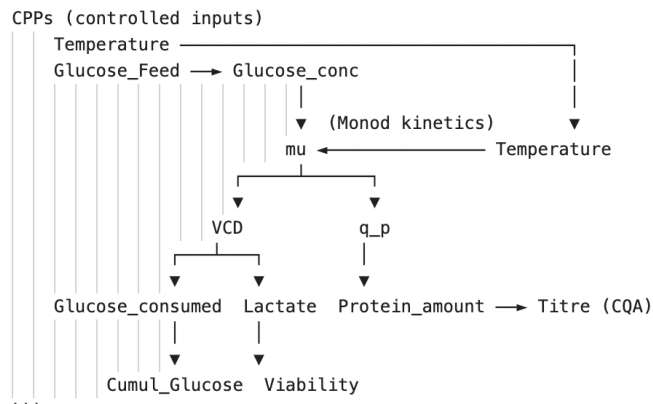


Figure 1 DAG design concept

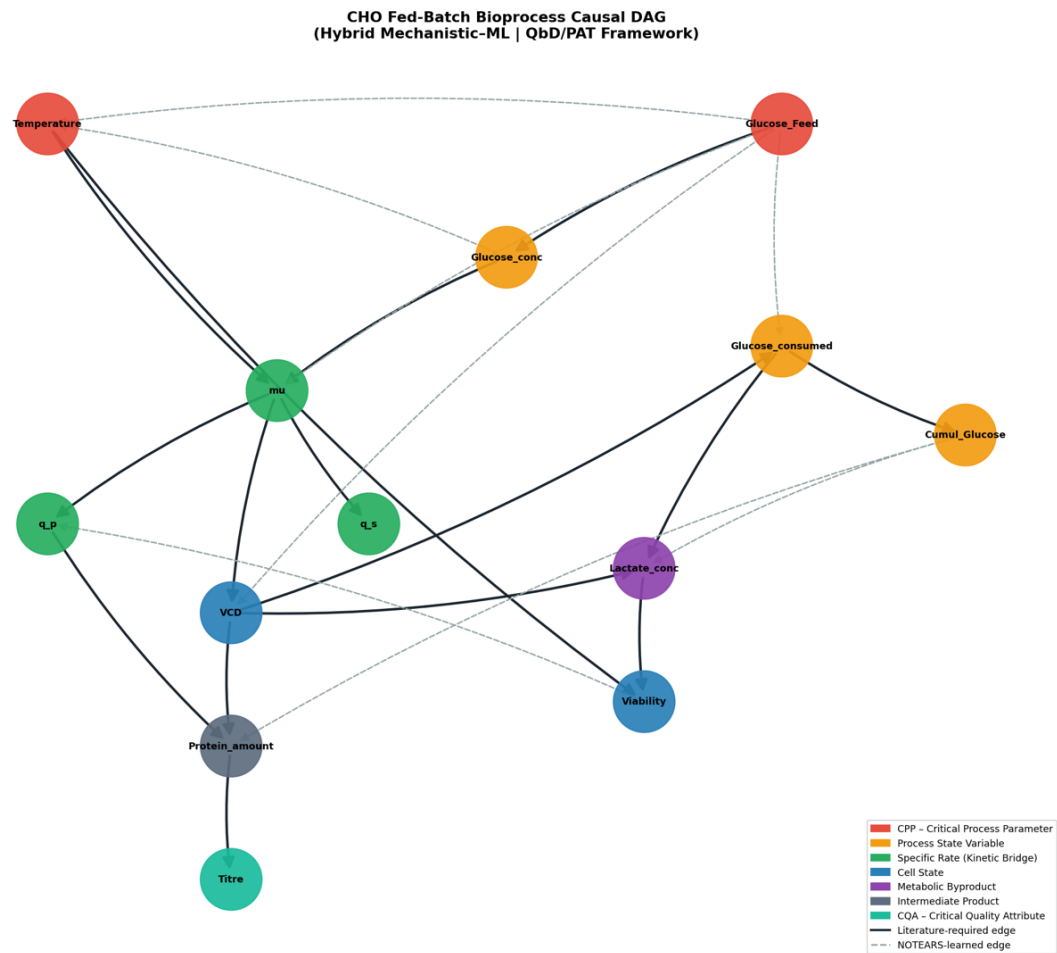


Figure 2 Causal Directed Acyclic Graph

3.2 Mechanistic equation model layer

The mechanistic layer estimates titer through a two-step kinetic model: Monod kinetics [7] for cell growth, and Luedeking-Piret (L-P) model [8][12] for product synthesis.

Step 1 — Cell Growth (Monod Kinetics)

The specific growth rate μ is defined as the normalized rate of change of viable cell density (VCD):

$$\mu = \frac{1}{X} \frac{dX}{dt} \text{ (unit: day}^{-1}\text{)}$$

where X denotes VCD and S denotes the glucose concentration. Under Monod kinetics, μ is a saturating function of substrate concentration:

$$\frac{dX}{dt} = \mu(S) \cdot X \quad (1)$$

$$\mu(S) = \frac{\mu_{\max} \cdot S}{K_s + S} \quad (2)$$

The parameters $\mu_{\max}$ (maximum specific growth rate) and K_s (half-saturation constant) are estimated by fitting equation (2) to observed (S, μ) pairs via equation (1).

Step 2 — Product Synthesis (Luedeking-Piret Model)

The specific productivity q_p is the product synthesis rate per cell, modelled as a linear combination of growth-associated (3) and non-growth-associated (4) terms [8]:

$$q_p = \alpha \cdot \mu + \beta \quad (3)$$

$$\frac{dP}{dt} = q_p \cdot X \quad (4)$$

where P denotes the cumulative product concentration, X is the viable cell density, and α and β are fitted from observed (μ, q_p) via equation (3).

Step 3 — Cumulative Titer

The mechanistic titer proxy P_{cum} is obtained by experimental data over time [5]:

$$P_{\text{cum}} = \sum_t q_p(t) \times X(t) \times \Delta t \quad (5)$$

P_cum is then mapped to the observed titer scale via Ridge regression, which absorbs any normalization constants introduced during data preprocessing. The subtract of mech_titer and actual_titer

$$\Delta T = T_{\text{mech}} - T_{\text{actual}}$$

Will further serves as the input data for the hybrid ML layer in 3.3.2.

Figure 3 and figure 4 shows the parameter values from mechanistic equations, which reveal that mechanistic has low R^2 in specific growth rate μ ($R^2 = 0.3221$) and specific productivity ($R^2 = 0.0023$). However, in figure 5 titer shows that mechanistic equation can efficiently predict titer, therefore it is assumed that it's more suitable for the hybrid-ML model compared to other critical quality attributes (CQAs). Three CQAs (μ , q_p , titer) will further be evaluated in the following hybrid model.

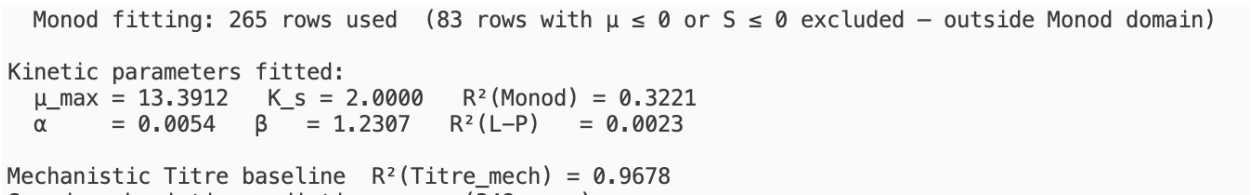


Figure 3 Result of mechanistic equation

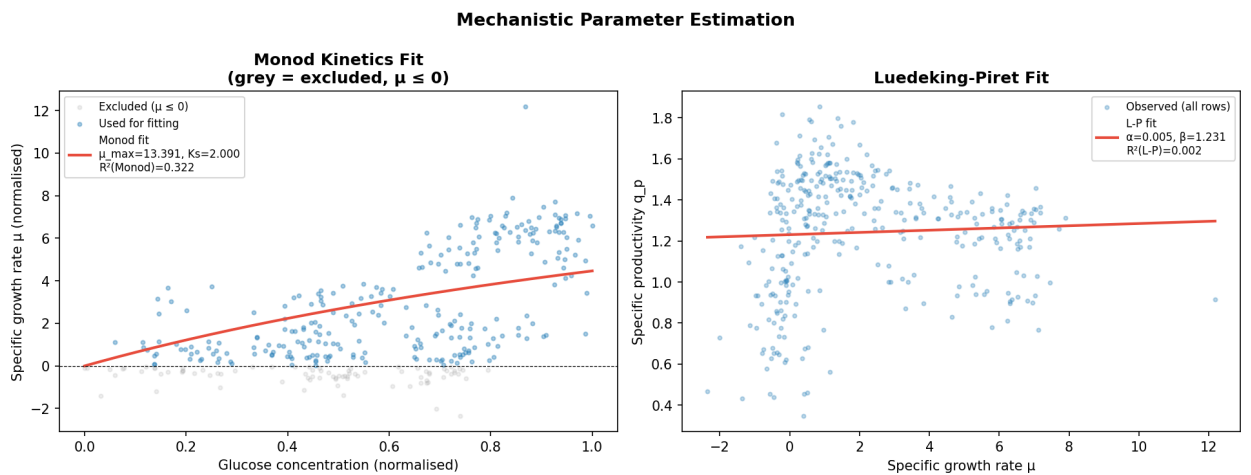


Figure 4 Mechanistic model for μ and q_p

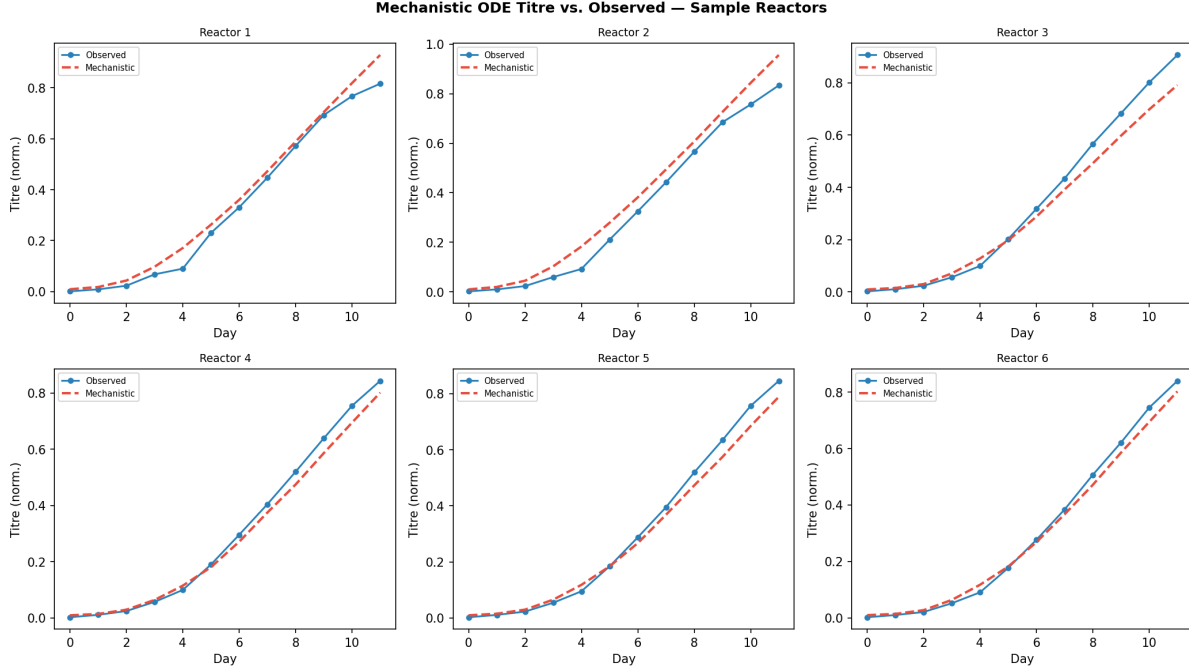


Figure 5 mechanistic model for titer

3.3.2 XGBoost titer residual prediction

The XGBoost titer residual prediction leverages XGBoost, which has been noted as one of the most efficient models for titer prediction [5]. Rather than predicting titer directly, the model is trained to predict the mechanistic residual — the deviation between the observed titer and the mechanistic baseline — such that the final hybrid prediction is:

$$\hat{T}_{\text{hybrid}} = T_{\text{mech}} + \hat{\epsilon}_{\text{XGB}} \quad (6)$$

where T_{mech} is the cumulative ODE-based titer from equation (5), and $\hat{\epsilon}_{\text{XGB}}$ is the residual predicted by XGBoost. This formulation ensures that the mechanistic structure is preserved while allowing the ML layer to correct systematic deviations that the ODE cannot capture.[5][6]

Feature Selection via Causal DAG

The feature set is restricted to the seven causal ancestors of Titer identified in the DAG: μ , q_p , VCD, Glucose_conc, Lactate_conc, Viability, and Temperature. This prevents bias from inputting the downstream feature and ensures that each predictor has a mechanistically justified pathway to titer. [4]

Hyperparameters

The XGBoost model is configured with $n_{\text{estimators}} = 400$, $\text{max_depth} = 4$, $\text{learning_rate} = 0.04$, and $\gamma = 0.1$. These settings balance the challenge of small dataset (29 bioreactors), avoiding the concerns of overfitting.[5]

3.3.3 Hybrid Model layer

The hybrid model combines the mechanistic ODE baseline with the XGBoost residual correction into a single prediction equation: [5][6]

$$\hat{T}_{\text{hybrid}} = T_{\text{mech}} + \hat{\varepsilon}_{\text{XGB}} \quad (7)$$

where T_{mech} is the mechanistic term derived from equation 5, and $\hat{\varepsilon}_{\text{XGB}}$ is the residual predicted by XGBoost from the seven DAG-identified features. Rather than learning the full titer signal from scratch, the XGBoost model is trained only on the difference between the observed titer and the mechanistic baseline from the prior step. The mechanistic layer anchors the prediction with physically consistent behavior, while the ML layer captures the biological variability that the ODE equations cannot explain. [1][6]

4. Model evaluation

4.1 Titer prediction with Hybrid model v.s. Pure XGBoost model

We evaluated the model with 5-fold validation, comparing the prediction performance of hybrid model and pure XGBoost model. The figure 6A shows that the hybrid-ML model has larger R2 (0.9730) than pure XGBoost (R2 = 0.9688), demonstrating the beneficial explanation of the hybrid model. Also, figure 6B shows that the titer predicted from hybrid model has a closer pattern to the observed titer. In figure 6C, the hybrid model overall shows more 0-oriented residual which means the model has smaller residual for most of the samples compares to raw mechanistic yield prediction.

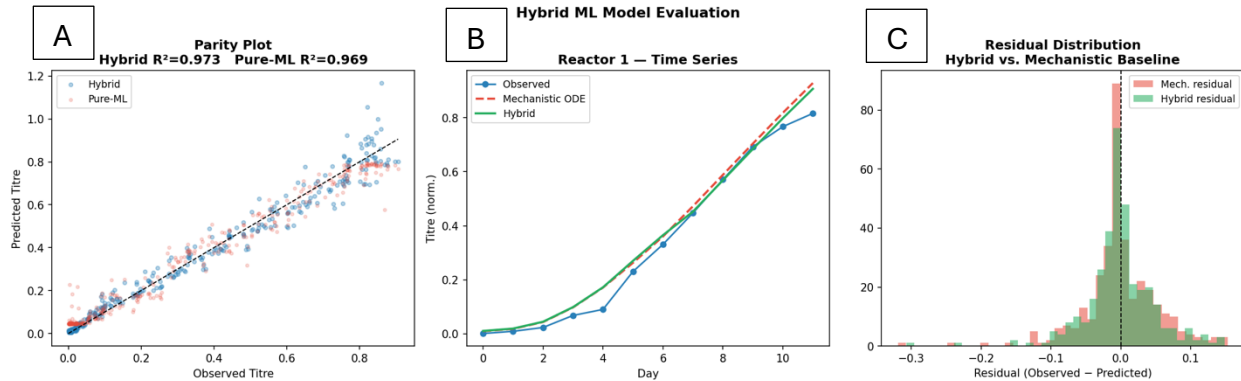


Figure 6 Performance comparison between hybrid-ML and pure-ML

4.2 μ and q_p prediction with Hybrid model v.s. Pure XGBoost model

Table 2 listed the R^2 for μ , q_p , and titer of using mechanistic, pure-ML, and hybrid-ML model. It is seen the mechanistic model poorly predicts μ and q_p ($R^2 = 0.2013$ and $R^2 = -0.0029$), therefore hybrid-ML—combining mechanistic and ML -- cannot outperform the pure ML. However, for titer, it mechanistic model performs well ($R^2 = 0.9678$), therefore integrating ML can efficiently improve the performance ($R^2 = 0.9730$) of the model by capturing residual pattern from the ML model.

Table 2 R^2 for 3 prediction models

Variable	ODE R^2	Pure-ML R^2	Hybrid R^2
μ	0.2013	0.8934	0.8911
q _p	-0.0029	0.7181	0.7162
Titre	0.9678	0.9688	0.9730

5. Discussion

This study has several limitations worth acknowledging. With only 29 bioreactors, the dataset is small, and there is a real risk that the XGBoost layer has learned reactor-specific patterns rather than generalizable bioprocess behavior. The hybrid model also depends heavily on the mechanistic baseline being reasonable — the Monod and Luedeking-Piret fits here are weak ($R^2 = 0.201$ and 0.002 respectively), which means the residual the ML layer is correcting is large and partially structured, not purely random noise. Testing on a different CHO dataset would be needed to confirm the results hold more broadly. The DAG structure is also worth treating with caution — NOTEARS identifies statistical associations in the data, and while the literature-enforced edges provide mechanistic grounding, the learned edges should not be interpreted as confirmed causal relationships without further experimental validation.

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